

第9讲. 泰勒公式(续). 极值

作业: P. 95. 3(1), (3), 4, 5, 6(1), (4)

复习: 设 $D \subset \mathbb{R}^2$ 是凸区域. $f(x, y) \in C^n(D)$. $P_0(x_0, y_0) \in D$. $\forall (x, y) \in D$. $f(x, y)$ 在 P_0 点可展开为带有拉格朗日余项的 n 阶泰勒展式

$$f(x, y) = f(x_0, y_0) + (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}) f(x_0, y_0) + \dots + \frac{1}{n!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(x_0, y_0) \\ + \frac{1}{(n+1)!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^{n+1} f(x_0 + \theta h, y_0 + \theta k) \quad 0 < \theta < 1.$$

$$h = x - x_0, \quad k = y - y_0.$$

定理1. 设 $D \subset \mathbb{R}^2$ 凸区域. $f(x, y) \in C^1(D)$. $P_0(x_0, y_0) \in D$. $\forall (x, y) \in D$.

$$f(x, y) - f(x_0, y_0) = f'_x(x_0 + \theta h, y_0 + \theta k) h + f'_y(x_0 + \theta h, y_0 + \theta k) k$$

$$h = x - x_0, \quad k = y - y_0.$$

(二元函数微分中值定理)

推论1. 设 $D \subset \mathbb{R}^2$ 凸区域且 $f(x, y) \in C^1(D)$. 且 $\text{grad} f(x, y) = 0$. ($\forall (x, y) \in D$)则 $f(x, y)$ 在 D 上常数.练习1. 设 $D \subset \mathbb{R}^2$ 凸区域. 且 $f(x, y) \in C^1(D)$. 且 $\vec{v}_1, \vec{v}_2$ 是线性无关向量.若 $\forall (x, y) \in D$. $\frac{\partial f}{\partial \vec{v}_i}(x, y) = 0$. ($i=1, 2$). 证明: $f(x, y)$ 在 D 上常数.定理2. 设 $D \subset \mathbb{R}^2$ 凸区域. $P_0(x_0, y_0) \in D$. $f(x, y) \in C^n(D)$.则 $\forall (x, y) \in D$. 当 $(x, y) \rightarrow (x_0, y_0)$ 时. $f(x, y)$ 在 P_0 点展开为带有皮亚诺余项的 n 阶泰勒展式

$$f(x, y) = \sum_{k=0}^n (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^k f(x_0, y_0) + o(\sqrt{h^2 + k^2}^n)$$

$$h = x - x_0, \quad k = y - y_0. \quad (h^2 + k^2)^{\frac{n}{2}}$$

证: $\forall (x, y) \in D$. $h = x - x_0, \quad k = y - y_0$.

$$f(x, y) = \sum_{k=0}^{n-1} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^k f(x_0, y_0) + \frac{1}{n!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(x_0 + \theta h, y_0 + \theta k), \quad 0 < \theta < 1.$$

||

$$\sum_{i=0}^n C_n^i \frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x_0+0h, y_0+0k) h^i k^{n-i}$$

$\parallel$
 $(h, k) \rightarrow (0, 0)$

$$\left(\frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x_0, y_0) + o(1) \right) h^i k^{n-i}$$

即证: $\underline{o(1)}(h^i k^{n-i}) = o((h^2+k^2)^{\frac{n}{2}}) \quad (h, k) \rightarrow (0, 0)$

$$h^i \leq (h^2+k^2)^{\frac{i}{2}}$$

$$k^{n-i} \leq (h^2+k^2)^{\frac{n-i}{2}}$$

$$h^i k^{n-i} \leq (h^2+k^2)^{\frac{n}{2}}$$

= . 极值 .

1. 无条件极值 .

定义 1. 设 $f(x, y)$ 在 (x_0, y_0) 的邻域 W 内有定义 .

若 $\forall (x, y) \in W, f(x, y) \leq f(x_0, y_0)$, 则称 $f(x, y)$ 在 (x_0, y_0) 取极大值 .

($\forall (x, y) \in W \setminus \{x_0, y_0\}, f(x, y) < f(x_0, y_0)$ 严格极大值)

称 $f(x_0, y_0)$ 为 $f(x, y)$ 极大值 . 且 (x_0, y_0) 为 $f(x, y)$ 极大值点 .

(. 严格极大值, 严格极大值点 .)

注: 极值局部 .

定义 2. 设 $f(x, y)$ 在 (x_0, y_0) 点存在偏导数 . 若 $f'_x(x_0, y_0) = 0, f'_y(x_0, y_0) = 0$.

称 (x_0, y_0) 为 $f(x, y)$ 的驻点或临界点 .

$$f(x, y) = \sqrt{x^2+y^2} \quad (0, 0) .$$

定理 3. 设 $f(x, y)$ 在 (x_0, y_0) 取得极值 . 若 $f(x, y)$ 在 (x_0, y_0) 存在偏导数, 则

$$f'_x(x_0, y_0) = 0, \quad f'_y(x_0, y_0) = 0 .$$

$f'_x(x_0, y_0)$ 存在 $f(x, y_0)$ 在 x_0 可导 且 x_0 取极值 .

$$f(x, y) \in C^2(D) \quad f_{xy}'' = f_{yx}''.$$

$$\forall P_0(x_0, y_0) \in D.$$

$$H_f(P_0) = \begin{pmatrix} f_{xx}''(P_0) & f_{xy}''(P_0) \\ f_{xy}''(P_0) & f_{yy}''(P_0) \end{pmatrix} \quad \text{实对称矩阵.}$$

→ f 在 P_0 的 Hesse 矩阵

$$\text{令 } \begin{pmatrix} x \\ y \end{pmatrix} = X$$

$$\textcircled{1} \text{ 若 对 } \forall X, \quad X^T H_f(P_0) X > 0. \quad \text{则 } H_f(P_0) \text{ 正定.}$$

$$\text{即 } \underbrace{f_{xx}''(P_0) > 0, \quad f_{xx}''(P_0) f_{yy}''(P_0) - f_{xy}''(P_0)^2 > 0.}$$

$$\textcircled{2} \text{ 若 } \forall X \neq 0, \quad X^T H_f(P_0) X < 0. \quad \text{则 } H_f(P_0) \text{ 负定.}$$

$$\text{即 } -H_f(P_0) \text{ 正定.}$$

$$\therefore \underbrace{f_{xx}''(P_0) < 0, \quad \text{且 } f_{xx}''(P_0) f_{yy}''(P_0) > f_{xy}''(P_0)^2.}$$

定理4. 设 $f(x, y)$ 在 $P_0(x_0, y_0)$ 的邻域 W 中有定义, 且 $f(x, y) \in C^2(W)$.

$$\text{且 } \text{grad } f(P_0) = 0. \quad \text{则}$$

① 当 $H_f(P_0)$ 正定, 则 P_0 为 $f(x, y)$ 严格极小值点.

② 当 $H_f(P_0)$ 负定, 则 P_0 为 $f(x, y)$ 严格极大值点.

③ 若 $H_f(P_0)$ 不定, 则 P_0 不是 $f(x, y)$ 极值点.

④ 若 $H_f(P_0)$ 半定, 则无法判断 P_0 是否为 $f(x, y)$ 极值点.

$$\begin{cases} f(x, y) = x^2 y^2 \\ f(x, y) = -x^2 y^2 \\ f(x, y) = x^2 y^3 \end{cases}$$

$$\text{证明: } \forall (x, y) \in W, \quad (x, y) \rightarrow (x_0, y_0).$$

$$\underline{f(x,y) - f(x_0,y_0)} = \frac{1}{2} \underbrace{(h \frac{\partial^2}{\partial x^2} + k \frac{\partial^2}{\partial y^2})^2 f(x_0,y_0)} + o(h^2+k^2)$$

$$h = x - x_0, \quad k = y - y_0$$

$$\frac{1}{2} \left[f''_{xx}(p_0) h^2 + 2 f''_{xy}(p_0) h k + f''_{yy}(p_0) k^2 \right]$$

||

$$\frac{1}{2} (h, k) H_f(p_0) \begin{pmatrix} h \\ k \end{pmatrix}$$

$$\Delta x = \begin{pmatrix} h \\ k \end{pmatrix}$$

||

$$\frac{1}{2} \Delta x^T H_f(p_0) \Delta x$$

① 若 $H_f(p_0)$ 正定, $\forall \Delta x \neq 0, \quad \Delta x^T H_f(p_0) \Delta x > 0.$

$\forall (x,y), \quad f(x,y) > \underline{f(x_0,y_0)}.$

② 若 $H_f(p_0)$ 负定, $\forall \Delta x \neq 0, \quad \Delta x^T H_f(p_0) \Delta x < 0.$

$\therefore f(x,y) < f(x_0,y_0).$

③ 若 $H_f(p_0)$ 不定, $\exists \Delta x \neq 0, \text{ s.t. } \Delta x^T H_f(p_0) \Delta x > 0 \Rightarrow f(x,y) > f(x_0,y_0)$
 $\exists \Delta y \neq 0, \text{ s.t. } \Delta y^T H_f(p_0) \Delta y < 0 \Rightarrow f(x,y) < f(x_0,y_0)$

定理 5. 设 $f(x,y)$ 在 $p_0(x_0,y_0)$ 邻域 W 内有定义, 且 $f(x,y) \in C^2(W)$, 且 $\text{grad} f(p_0) = 0$.

① 若 $\exists \delta > 0$, s.t. $\forall p(x,y) \in \mathcal{B}(p_0, \delta) \subset W$, 有 $H_f(p)$ 正定, 则 p_0 为 $f(x,y)$ 极小值点.

② 若 $\exists \delta > 0$, s.t. $\forall p(x,y) \in \mathcal{B}(p_0, \delta) \subset W$, 有 $H_f(p)$ 半负定, 则 p_0 为 $f(x,y)$ 极大值点.

证: $\forall p(x,y) \in W, \quad p^*$ 位于 $\overline{p_0 p}$ 上一点.

$$f(x,y) - f(x_0,y_0) = \frac{1}{2} \left(f''_{xx}(p^*) h^2 + 2 f''_{xy}(p^*) h k + f''_{yy}(p^*) k^2 \right)$$

$$= \frac{1}{2} \Delta x^T H_f(p^*) \Delta x \geq 0.$$

$$\Delta x = \begin{pmatrix} h \\ k \end{pmatrix}$$

例1. $f(x,y) = x^3 + e^{3y} - 3xe^y$. 讨论其极值. 并求值.

解:
$$\begin{cases} f'_x(x,y) = 3x^2 - 3e^y = 0 \Rightarrow x^2 = e^y \\ f'_y(x,y) = 3e^{3y} - 3xe^y = 0 \Rightarrow e^{2y} = x \end{cases} \Rightarrow e^{4y} = e^y \Rightarrow e^{3y} = 1$$

$$\Rightarrow \begin{cases} y=0 \\ x=1 \end{cases}$$

驻点 $P(1,0)$

$$f(-1,0) = -1 < -1$$

$$f''_{xx}(x,y) = 6x$$

$$f''_{xx}(1,0) = 6 > 0$$

$$f(1,0) = -1$$

$$f''_{xy}(x,y) = -3e^y$$

$$f''_{xy}(1,0) = -3$$

$$f''_{yy}(x,y) = 9e^{3y} - 3xe^y$$

$$f''_{yy}(1,0) = 9 - 3 = 6$$

$$36 - 9 > 0$$

$H_f(P)$

$H_f(P)$ 正定.

$$H_f(P) = \begin{pmatrix} 6x & -3e^y \\ -3e^y & 9e^{3y} - 3xe^y \end{pmatrix} \text{ 不定.}$$

例2. 设 $f(x,y) \in C^2$. $g(x,y) = f(e^{xy}, x^2+y^2)$ 且

$$f(x,y) = 1 - x - y + o(\sqrt{(x-1)^2 + y^2}) \quad ((x,y) \rightarrow (1,0))$$

证明: $g(x,y)$ 在 $(0,0)$ 取得极值. 判断此极值是极大值或极小值. 并求极值.

分析. ① 证: $g'_x(0,0) = 0$. $g'_y(0,0) = 0$.

证明:

$$\therefore f(x,y) = -(x-1) - y + o(\sqrt{(x-1)^2 + y^2}) \quad (x,y) \rightarrow (1,0)$$

$$\therefore f(1,0) = 0, \quad f'_x(1,0) = -1, \quad f'_y(1,0) = -1$$

$$g'_x(x,y) = f'_1(e^{xy}, x^2+y^2) \cdot y e^{xy} + f'_2(e^{xy}, x^2+y^2) \cdot 2x$$

$$g'_y(x,y)$$